

A comparison of downscaled and raw GCM output: implications for climate change scenarios in the San Juan River basin, Colorado

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Abstract

The fundamental rationale for statistical downscaling is that the raw outputs of climate change experiments from General Circulation Models (GCMs) are an inadequate basis for assessing the effects of climate change on land-surface processes at regional scales. This is because the spatial resolution of GCMs is too coarse to resolve important sub-grid scale processes (most notably those pertaining to the hydrological cycle) and because GCM output is often unreliable at individual and sub-grid box scales. By establishing empirical relationships between grid-box scale circulation indices (such as atmospheric vorticity and divergence) and sub-grid scale surface predictands (such as precipitation), statistical downscaling has been proposed as a practical means of bridging this spatial difference. This study compared three sets of current and future rainfall-runoff scenarios. The scenarios were constructed using: (1) statistically downscaled GCM output; (2) raw GCM output; and (3) raw GCM output corrected for elevational biases. Atmospheric circulation indices and humidity variables were extracted from the output of the UK Meteorological Office coupled ocean-atmosphere GCM (HadCM2) in order to downscale daily precipitation and temperature series for the Animas River in the San Juan River basin, Colorado. Significant differences arose between the modelled snowpack and flow regimes of the three future climate scenarios. Overall, the raw GCM output suggests larger reductions in winter/spring snowpack and summer runoff than the downscaling, relative to current conditions. Further research is required to determine the generality of the water resource implications for other regions, GCM outputs and downscaled scenarios. © 1999 Elsevier Science B.V. All rights reserved.

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1. Introduction

An often stated justification for statistical downscaling is that the raw output of climate change experiments from General Circulation Models (GCMs) are an inadequate basis for assessing land-surface impacts at regional scales (DOE, 1996). This is because the spatial resolution of GCM grids is too

coarse to resolve many important sub-grid scale processes (most notably those pertaining to the hydrological cycle) and because GCM output is often unreliable at individual grid and sub-grid box scales (IPCC, 1996). This mismatch, between what the climate impacts community requires and what the GCMs are able to supply, has been a confounding issue affecting the confidence placed in impact scenarios at the basin scale (Hostetler, 1994).

A wide variety of techniques exist for assessing the effects of climate change on water resources (see the

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overview by Leavesley (1994)). The most conventional solution to the problem is to perturb historical time series of high resolution meteorological variables by the difference (or ratio) of the means of GCM output between the altered and control climate runs (e.g. Arnell, 1996). An alternative methodology, statistical downscaling, involves bridging the two discordant scales by establishing empirical relations between features reliably simulated by the GCM at grid-box scales (such as geopotential height fields) and surface predictands at sub-grid scales (such as precipitation occurrence or amounts). These procedures are analogous to the so-called “Model Output Statistics” (MOS) and “Perfect Prog” (PP) techniques used since the 1970s for short range numerical weather prediction (Klein and Glahn, 1974).

Although many studies have discussed the theory and practice of statistical downscaling (e.g. Kim et al., 1984; Karl et al., 1990; Wigley et al., 1990; Bardossy and Plate, 1992; Hay et al., 1992; von Storch et al., 1993), relatively few have explicitly considered the limitations of such techniques (Giorgi and Mearns, 1991; Wilby and Wigley, 1997). However, sensitivity analyses have demonstrated the susceptibility of downscaled scenarios to season definitions, the choice of data standardisation technique, length of calibration period, function form and predictor variable(s) (e.g. Winkler et al., 1997). It has also been shown that different circulation schemes (Buishand and Brandsma, 1997) and downscaling methodologies (Wilby et al., 1998b) yield markedly different regional climate change scenarios, even when common sets of GCM predictors are used. Finally, there is skepticism regarding the assumed stationarity of predictor–predictand relations (Pielke, personal communication; Wilby, 1997) and the reproduction of low-frequency surface climate variability in downscaling schemes continues to be problematic (Katz and Parlange, 1996). Although it is not our intention to investigate such assumptions, it is appropriate at the outset to flag sources of uncertainty that are inherent to most statistical downscaling approaches.

Given the aforementioned limitations, it is important that the relative merits of downscaled and raw GCM output should be properly compared. Although there have been numerous validations of the synoptic circulation patterns and climate variables produced by GCMs (e.g. Santer and Wigley, 1990; McCabe and

Legates, 1992; Hulme et al., 1993; Airey and Hulme, 1995; McKendry et al., 1995; Osborn and Hulme, 1998), comparatively little is known about the “value-added” (or indeed “value-subtracted”) of downscaled versus raw GCM output, especially when applied to (non-linear) impact models. In other words, to what extent do downscaled scenarios (resulting from a range of subjective methodological decisions and driven by imperfect GCM predictors) actually yield regional climate change impacts that are significantly different to those derived from raw GCM output?

With this question in mind—as well as the additional effort required to generate downscaled climate scenarios—the present study compares three sets of current and future daily rainfall-runoff scenarios. These scenarios were constructed using: (1) statistically downscaled GCM output; (2) raw GCM output; and (3) raw GCM output corrected for elevational biases. Atmospheric circulation indices and humidity variables derived from the UK Meteorological Office coupled ocean–atmosphere GCM (Johns et al., 1997; Mitchell and Johns, 1997) were used to downscale daily precipitation and temperature series for the Animas River, a sub-basin of the San Juan River, Colorado (see Fig. 1). The Animas River basin has a drainage area of 1820 km² and elevation that ranges from approximately 2000 to 4000 m. The climate scenarios generated were then used to drive a distributed hydrological model. Changes in the modelled daily flow regime and snowpack behaviour between current and future climate scenarios were compared. Finally and in the light of these results, we discuss the relative merits of the three techniques for regional scenario development and climate change impact assessment.

2. Data

Three data sets were compiled for the purposes of climate scenario generation and hydrological model application. Data sets were compiled on a water-year (WY) basis with October 1980 to September 1981 being WY 1981.

2.1. Station data

Daily maximum and minimum temperatures

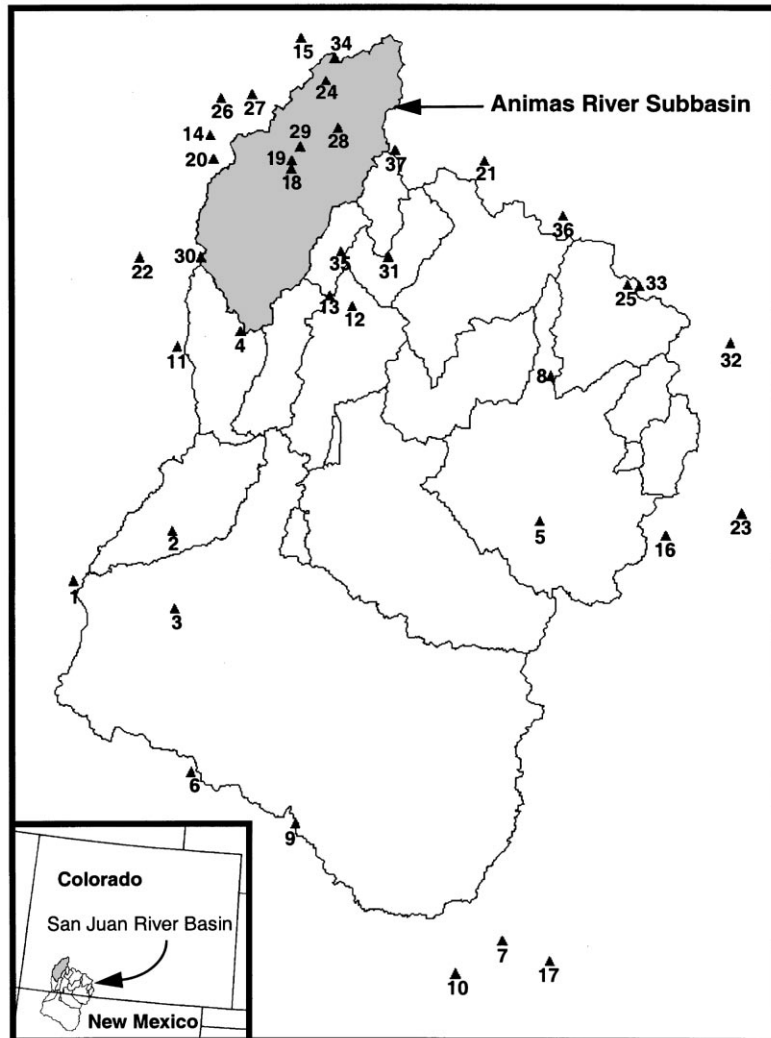


Fig. 1. Location map showing the meteorological station network of the San Juan River basin and Animas River sub-basin, Colorado.

(TMAX and TMIN) and precipitation amounts (PRCP) from 37 stations in and around the San Juan River basin were compiled (see Fig. 1 and Table 1). Twenty-two of the precipitation stations are relatively high elevation (2500–3500 m) Snow Telemetry (SNOTEL) stations. The remainder of the stations are National Weather Service (NWS) stations.

2.2. Re-analysis data

The ultimate choice of predictors for downscaling is constrained by three main factors. The predictor

variables should be (1) reliably simulated by the GCM under consideration, (2) readily available from (in this case, daily) archives of GCM output and (3) strongly correlated with the surface variable(s) of interest. Using these criteria, daily grid point data for mean sea level pressure (mslp), 500 hPa geopotential heights (H), 2 m (near surface) temperatures (T_{2m}) and 0.995 sigma level (near surface) relative humidities (RH) were obtained from the National Center for Environmental Prediction (NCEP) re-analysis (Kalnay et al., 1996) for the period WY 1981–1995. All data were re-gridded from the NCEP

Table 1

Station details for the San Juan River basin (the stations marked with ‘*’ were selected for the Animas River sub-basin downscaling; all stations were used to calculate area average PRCP, TMAX and TMIN for the San Juan River basin—a domain equating to a typical HadCM2 grid-box area)

Station type	Station number	Station name	Elevation (m)	Start of record
National Weather Service Station	1	Farmington Experimental Station	1716	1948
	2	Aztec Ruins National Monument	1719	1948
	3	Bloomfield	1771	1948
	4	Durango	2012	1948
	5	Dulce	2070	1948
	6	Otis	2097	1957
	7	Cuba	2149	1948
	8	Pagosa Springs	2167	1948
	9	Lybrook	2179	1951
	10	Johnson Ranch	2195	1948
	11	Fort Lewis	2316	1948
	12*	Vallecito Dam	2332	1948
	13	Lemon Dam	2466	1982
	14	Rico	2676	1948
Snow Telemetry Station SNOTEL	15	Telluride	2682	1948
	16	Chamita	2560	1979
	17	Senorita Divide	2621	1980
	18	Cascade	2707	1978
	19	Cascade 2	2719	1990
	20	Scotch Creek	2774	1985
	21	Upper Rio Grande	2865	1986
	22	Mancos	3048	1994
	23	Cumbers Trestle	3054	1979
	24*	Mineral Creek	3060	1978
	25	Upper San Juan	3088	1978
	26	El Diente Peak	3109	1985
	27	Lizard Head Pass	3109	1979
	28	Molas Lake	3200	1985
	29	Spud Mountain	3249	1986
	30	Columbus Basin	3287	1994
	31	Vallecito	3316	1986
	32	Lily Pond	3353	1979
	33*	Wolf Creek Summit	3353	1986
	34	Red Mountain Pass	3399	1980
	35	Stump Lakes	3414	1986
	36	Middle Creek	3429	1979
	37	Beartown	3536	1982

grid (1.875° latitude by 1.875° longitude) to the GCM grid (2.5° latitude by 3.75° longitude).

The mslp and H data were used to calculate four daily airflow indices for the surface and upper atmosphere, respectively, according to the methodology described by Jones et al. (1993). At each atmospheric level the derived circulation indices were: zonal and meridional components of the geostrophic airflow, U and V ; total shear vorticity, Z ; and divergence, D . A positive value of U is indicative of an airflow from

west to east and positive V corresponds to an airflow from south to north. The vorticity (Z) is a measure of atmospheric rotation with negative values indicative of anticyclonic (high pressure weather) and positive values corresponding to cyclonic (low pressure weather) conditions. The divergence (D) is a measure of atmospheric motion in the vertical plane, being positive when air flows diverge and negative when convergent.

Daily mean temperatures and relative humidities at

Table 2
Candidate predictor variables (* denotes variables used in downscaling)

Predictor variable	Abbreviation	Source
<i>Surface variables</i>		
*Mean sea level pressure	mslp	NCEP
Zonal velocity component	Us	Derived from mslp
Meridional velocity component	Vs	Derived from mslp
Strength of the resultant flow (hPa)	Fs	Derived from mslp
Vorticity (hPa)	Zs	Derived from mslp
Divergence (hPa)	Ds	Derived from mslp
2 m temperatures (°C)	T2m	NCEP
Relative humidities (%)	RH	NCEP
*Specific humidity (gm/kg)	SH	Derived from RH and T2m
<i>Upper-atmosphere variables (500 hPa)</i>		
*500 hPa geopotential heights (m)	H	NCEP
Zonal velocity component	Uu	Derived from H
Meridional velocity component	Vu	Derived from H
Strength of the resultant flow (hPa)	Fu	Derived from H
Vorticity (hPa)	Zu	Derived from H
Divergence (hPa)	Du	Derived from H

the surface were used to estimate daily mean specific humidities via the non-linear approximation of Richards (1971). Finally, daily values of all 15 candidate variables (Table 2) were extracted from the global fields for the GCM grid-box (2084) nearest to the San Juan River basin.

2.3. General circulation model output

The GCM used was the UK Meteorological Office, Hadley Centre's coupled ocean/atmosphere model (HadCM2) forced by combined CO₂ and albedo (as a proxy for sulphate aerosol) changes (Johns et al., 1997; Mitchell and Johns, 1997). In this 'SUL' (sulphate-plus-greenhouse gas) experiment, the model run begins in 1861 and is forced with an estimate of historical forcing to 1990 and a projected future forcing scenario over 1990–2100. The historical forcing is only an approximation of the "true" forcing so the GCM results for WY 1981–1995 would not be expected to exactly represent present-day conditions (for more details see Wilby et al., 1998b, Appendix A), nor are the GCM years directly equivalent to actual years due to the difference in observed climate and GCM forcing (see Wilby et al., 1998b). With these caveats in mind, HadCM2 output for the period WY 1981–1995 was used as the best available proxy for the present climate as in previous downscaling studies (e.g. Conway et al., 1996; Pilling et al.,

1998; Wilby et al., 1998a,b). With the exception of daily mslp, the normalised predictor variables produced by HadCM2 for the years 1981–1995 were statistically indistinguishable ($P < 0.05$) from those of NCEP for the same period. The discrepancy for mslp was attributed, in part, to missing data in the HadCM2 archive for all Januarys between 1981 and 1990 (totalling 300 days).

Two time series of daily mean sea level pressure, 500 hPa geopotential heights, surface relative humidity, maximum and minimum temperatures were obtained from the HadCM2 'SUL' experiment. The first set of data, representative of the current (WY 1981–1995) climate, parallels the NCEP re-analysis data; the second, represents future (WY 2081–2095) climate conditions due to anthropogenic forcing. Both 15 year data sets were used to derive the chosen predictor variables for the statistical downscaling (Table 2). Finally, daily PRCP, TMAX and TMIN for the HadCM2 grid-box (2084) were retained for both time periods in order to simulate changes in the daily flows of the Animas River basin using the raw GCM scenarios.

3. Methodology

The compiled data sets were used to develop six current and three future climate scenarios to

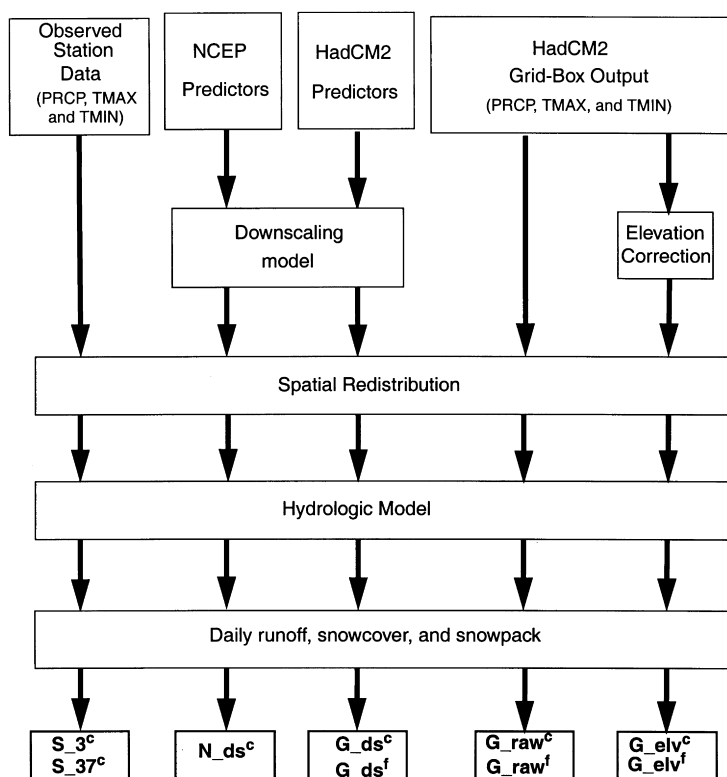


Fig. 2. Methods for generation of current (c) and future (f) rainfall-runoff scenarios for the Animas River basin. (S_3^c uses three stations and S_{37}^c uses 37 stations).

investigate the raw GCM and downscaled GCM methods of scenario generation in the Animas River basin. The description of the methodologies involved will follow the routes shown in Fig. 2. Daily precipitation and temperature time series produced from observed, downscaled and raw GCM output were distributed spatially over the Animas River basin using monthly lapse rates calculated from observed data. These spatially distributed hydrometeorological variables were then used as input to the watershed model PRMS (Precipitation-Runoff Modelling System) (Leavesley et al., 1983; Leavesley and Stannard, 1995). Finally, modelled daily flows, snow-covered area and snowpack water equivalents were used as measures to evaluate the scenarios and future climate change impact. The following sections provide details of the methodologies involved at each stage of the analysis.

3.1. Climate scenario definitions

The S_3^c and S_{37}^c series correspond to observed station data for current (c) conditions in the Animas River basin (three stations) and entire San Juan River basin (37 stations), respectively. S_3^c data were used to calibrate the downscaling model and S_{37}^c were used to compare observed climate conditions with GCM output for the current climate (i.e. the entire San Juan River basin area of 37,500 km² equates to a single HadCM2 grid-box). N_{ds}^c (Fig. 2) represents the downscaling model estimate of current observed conditions in the Animas River basin given only grid-box values of the chosen NCEP predictor variables (see later). G_{ds}^c and G_{ds}^f are the downscaled time-series produced for current and future (f) climate periods using GCM (HadCM2) output as the predictor variable source. G_{raw} and G_{elv} denote the raw and elevation bias corrected GCM output, respectively

(see Section 3.2.2). The G_{raw}^c , G_{raw}^f , G_{elv}^c and G_{elv}^f scenarios all use daily values of PRCP, TMAX and TMIN obtained from HadCM2 at the grid-point nearest to the San Juan River basin for the current and future climate.

3.2. Statistical downscaling model

The statistical downscaling model was calibrated using S_3^c data, originating from the three stations (denoted by ‘*’ in Table 1) which produced the best simulations of observed daily flows in the Animas River basin (see Section 3.2.2). The downscaling model parameters were estimated by linear least-squares regression using daily data for the nine WYs 1987–1995 and evaluated using the six WYs 1981–1986. Although data were available for previous years the station network was not as dense, contributing to likely underestimates of true area-average wet-day frequencies and precipitation totals (see Osborn and Hulme, 1997).

The downscaling model was calibrated using daily area-averaged series (S_3^c) of wet-day occurrence (O), wet-day amounts (R), maximum (TMAX) and minimum (TMIN) temperatures for WYs 1987–1995, with separate regressions for each climatological season, i.e. individual winter (DJF), spring (MAM), summer (JJA) and autumn (SON) sub-models. All daily predictor variables were first normalised using the corresponding period means and standard deviations (as advocated by Karl et al. (1990)). Three predictor variables were selected following a stepwise multiple linear regression analysis of the 15 candidate variables listed in Table 2. The chosen predictors were grid-box (HadCM2 number 2084) values of daily specific humidity (SH), mean sea level pressure (mslp) and 500 hPa geopotential heights (H). These three predictors (SH, mslp and H) were selected by the stepwise regression on 12/16, 11/16 and 15/16 occasions, respectively (i.e. 4 predictands $\times$ 4 seasons = 16 sub-models). Thus, it was possible to downscale all four surface predictands with a parsimonious, yet physically plausible, set of predictors.

The legitimacy of the regression analysis of daily $\ln(R)$, TMAX, TMIN, mslp, H and SH was verified by testing for normality. All variables were found to be approximately normal within the following percentile

ranges: $\ln(R)$ all values; TMAX 5–95%; TMIN <95%; mslp between 5 and 95%; SH 25–75%; and H < 90%. Overall, the assumption of normality held for all but the most extreme values of the predictors and predictands, with the most serious violation being for values of SH outside the inter-quartile range. Detailed descriptions of each of the downscaling model components are now provided.

3.2.1. Daily precipitation occurrence (O_i)

Daily probabilities of non-zero precipitation (a wet-day) O_i for a given day i were downscaled using the three grid-box predictor variables SH, mslp and H and a lag-1 autocorrelation parameter. The random variable O_i was modelled using the following regression equation:

$$O_i = \alpha_0 + \alpha_{O_{i-1}}O_{i-1} + \alpha_{\text{SH}}\text{SH}_i + \alpha_{\text{mslp}}\text{mslp}_i + \alpha_H H_i \quad (1)$$

The α parameters were estimated using linear least squares regression. A uniformly distributed random number r ($0 \leq r \leq 1$) was used to determine whether precipitation occurs. For a given site and day, a wet-day was returned if $r \leq O_i$.

3.2.2. Daily precipitation amounts (R_i)

If determined that precipitation has occurred, the daily precipitation amount was also downscaled using the three grid-box predictor variables SH, mslp and H . Since the wet-day precipitation amounts (R_i) for a given day i are always non-zero, it is appropriate to formulate the following regression model (following Kilsby et al., 1998):

$$R_i = \exp(\beta_0 + \beta_{\text{SH}}\text{SH}_i + \beta_{\text{mslp}}\text{mslp}_i + \beta_H H_i + \epsilon_i) \quad (2)$$

where the β 's were parameters estimated using linear least squares regression and ϵ_i was random or modelling error. The expected value was given by

$$E(R_i) = \phi_{c_R} \exp(\beta_0 + \beta_{\text{SH}}\text{SH}_i + \beta_{\text{mslp}}\text{mslp}_i + \beta_H H_i) \quad (3)$$

where c_R was an empirically derived correction ratio that allows for the bias resulting from the re-transformation of $\ln(R)$ to R and the fact that ϵ_i came from a skewed distribution. The value of c_R was constrained such that observed and downscaled precipitation

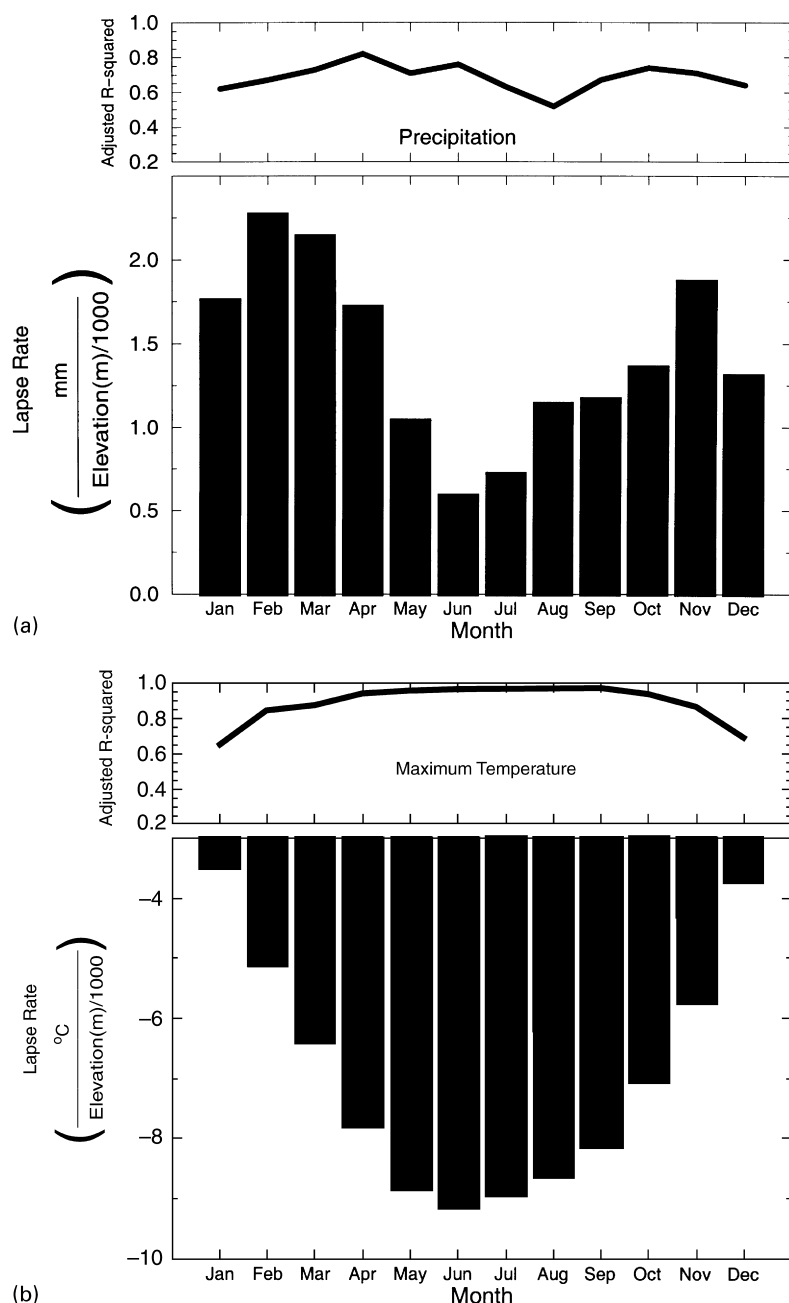


Fig. 3. Observed lapse rates by month for (a) precipitation, (b) maximum and (c) minimum temperatures in the San Juan River basin, 1988–1997.

totals were equal for the simulation period. A random scaling factor ϕ (with a mean of 1) was used to increase the variance of R to agree better with observations (as in Hay et al., 1991). Note that a lag-1

autoregressive component was not used to model R_i because its inclusion did not significantly improve the explained variance in wet-day amounts for the Animas River basin. However, it is acknowledged

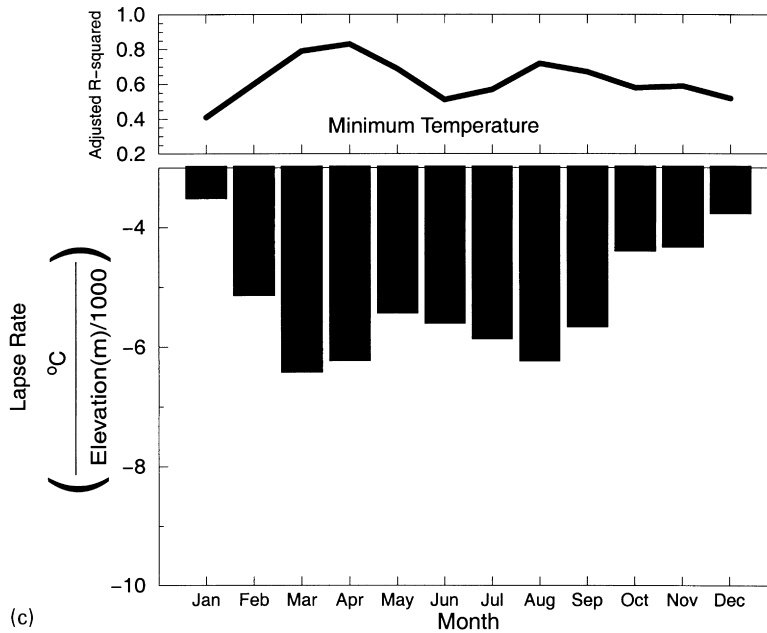


Fig. 3. (continued)

that the inclusion of this parameter may be appropriate at other locations.

3.2.3. Daily temperatures ($TMAX_i$ and $TMIN_i$)

Daily maximum ($TMAX_i$) and minimum ($TMIN_i$) temperatures for a given day i were downscaled using the three grid-box predictor variables SH , $mslp$ and H and the preceding days' maximum ($TMAX_{i-1}$) and minimum ($TMIN_{i-1}$) temperatures, respectively. The daily temperature series were modelled using the following regression equations:

$$TMAX_i = \delta_0 + \delta_{TMAX_{i-1}} TMAX_{i-1} + \delta_{SH} SH_i + \delta_{mslp} mslp_i + \delta_H H_i + \zeta_i \quad (4)$$

$$TMIN_i = \gamma_0 + \gamma_{TMIN_{i-1}} TMIN_{i-1} + \gamma_{SH} SH_i + \gamma_{mslp} mslp_i + \gamma_H H_i + \xi_i \quad (5)$$

where δ and γ were parameters estimated by linear least squares regression and ζ_i and ξ_i were random or modelling errors. Both ζ_i and ξ_i were assumed to be normally distributed with mean zero and standard deviation σ equal to the standard error of the regression equation. Both sets of residuals were

modelled stochastically using conventional Monte Carlo methods.

3.3. Spatial distribution and elevation correction schemes

Daily values of PRCP, TMAX and TMIN were distributed across the Animas River basin using a monthly lapse rate for each variable. Lapse rates were calculated using observed data from NWS and SNOTEL stations (see Table 1) for the period 1988–1997. A mean value of PRCP, TMAX and TMIN for these data stations and their corresponding mean elevations were computed daily. These daily mean values were then used with the monthly lapse rates to distribute PRCP, TMAX and TMIN according to the mean elevations of individual basin sub-areas delineated for hydrological modelling purposes. These hydrological response units (HRUs) are discussed in Section 3.4. Monthly lapse rates and corresponding adjusted R -squared values are shown in Fig. 3a–c.

A Monte Carlo analysis was used to determine the optimal number of stations for use in the distribution methodology. To initiate the Monte Carlo (MC) analysis, each station was tested individually for

distributing PRCP, TMAX and TMIN in the Animas River basin. Thirty-seven time-series of PRCP, TMAX and TMIN were calculated and used as input into the hydrological model (see later). In the preliminary screening, meteorological stations yielding the highest absolute errors between observed and simulated runoff for the period 1988–1997 (30% of the stations) were rejected.

Further MC testing using all possible combinations of two, three and four of the remaining stations was next conducted. For each set, the best station combination was determined. The MC analysis ended when the absolute error associated with the best station combination showed no significant improvement from one set to the next. In this exercise no improvement was seen by increasing from three to four stations. The optimal station set for the Animas River basin was determined to be the three stations: Vallecito Dam, Mineral Creek and Wolf Creek Summit (see Table 1). The selection of Wolf Creek Summit by the MC analysis highlights the problem of gauge under catch, estimated to be in the region of 20–50% for snowfall in mountainous terrain (Severuk, 1989). Note that no correction was made for gauge under catch in this study. The choice of Wolf Creek Pass compensates for gauge under catch since this station generally has precipitation amounts in the winter time that are 20% higher than those measured at stations with similar elevations in the San Juan River basin.

The three stations were used to construct the S_3^c scenario against which the downscaling model was calibrated and then used to generate the N_{ds}^c , G_{ds}^c and G_{ds}^f scenarios. Downscaled PRCP, TMAX and TMIN time-series data for present (N_{ds}^c , G_{ds}^c) and future (G_{ds}^f) conditions were then spatially distributed across the Animas River basin prior to input into the hydrological model. All downscaled time-series were treated as one station in the distribution methodology with a mean elevation of 2915 m (the mean elevation of the three stations chosen in the MC analysis). This information was used along with the monthly lapse rates to distribute PRCP, TMAX and TMIN to each HRU.

GCM grid-box PRCP, TMAX and TMIN time-series data were spatially distributed in the Animas River basin using two methods. The first method used the mean elevation of the GCM grid node

(1900 m) as the mean station elevation in the distribution methodology and was used to produce scenario G_{raw}^c and G_{raw}^f . The second method adjusts the GCM elevation value and was used to produce scenarios G_{elv}^c and G_{elv}^f . This was necessary because the GCM grid-box PRCP, TMAX and TMIN values were not representative of current climate conditions at 1900 m, but had representative elevations that varied by month and were all higher than 1900 m. In order to produce realistic estimates of current PRCP, TMAX and TMIN, the second method used the elevations shown in Fig. 4 to distribute these variables. These elevations were determined by calculating the elevation value needed to make monthly estimates of PRCP, TMAX and TMIN used in the G_{raw}^c scenario consistent with observed data. Compared to the first method, the use of higher representative elevations increased temperature (due to negative lapse rates) and decreased precipitation (due to positive lapse rates) over the basin.

3.4. Hydrological modelling

The hydrological model used was the US Geological Survey's (USGS) Precipitation-Runoff Modelling System (PRMS) (Leavesley et al., 1983; Leavesley and Stannard, 1995). PRMS is a distributed-parameter, physical process watershed model. Distributed-parameter capabilities are provided by partitioning a watershed into units, using characteristics such as slope, aspect, elevation, vegetation type, soil type and precipitation distribution. Each unit is assumed to be homogeneous with respect to its hydrologic response and to the characteristics listed above. Each unit is termed a hydrologic response unit (HRU). A water balance and an energy balance are computed daily for each HRU. The sum of the responses of all HRUs, weighted on a unit-area basis, produces the daily watershed response.

PRMS was applied using the USGS Modular Modeling System (MMS) (Leavesley et al., 1996). A major component of MMS is the GIS Weasel, a Geographic Information System (GIS) interface developed to provide a variety of GIS tools to delineate, characterise and parameterise the topographic and hydrologic features of a watershed. Using the GIS Weasel, the Animas River basin was delineated into 34 HRUs. Parameters for each HRU related to

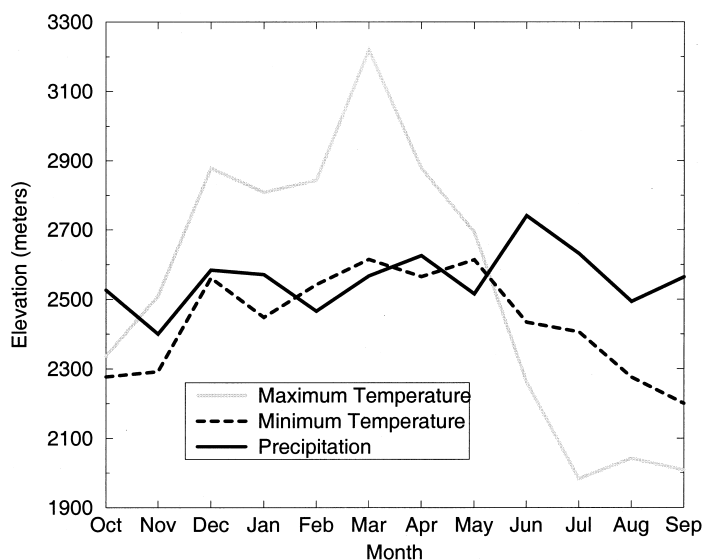


Fig. 4. Corrected elevations by month for HadCM2 daily precipitation, maximum and minimum temperatures.

topographic, vegetation and soils characteristics were computed using three digital databases: (1) USGS 3-arc second digital elevation models; (2) State Soils Geographic (STATSGO) 1 km gridded soils data (US Department of Agriculture, 1994); (3) Forest Service 1 km gridded vegetation type and density data (US Department of Agriculture, 1992). Other model parameters were estimated using model applications in comparable basins of this region (Leavesley et al., 1992). To prevent biasing parameter estimates to any particular meteorological data set, no parameter optimisation was performed.

Snow is the major form of precipitation input to the Animas River basin and the major source of streamflow. The snow components of PRMS simulate the accumulation and depletion of a snowpack on each HRU. A snowpack is maintained and modified both as a water reservoir and as a dynamic heat reservoir. A water balance is computed daily and an energy balance is computed twice each day. The energy-balance computations include net shortwave and longwave radiation, approximations of convection and condensation terms and the heat content of precipitation.

In summary, PRMS uses daily inputs of solar radiation and the variables PRCP, TMAX and TMIN. Solar radiation was distributed to each HRU as a function of HRU slope and aspect. Solar radiation data were not

available on a daily basis for either the measurement stations or the archived GCM output and so were computed using existing algorithms in PRMS. Estimates of daily shortwave radiation received on a horizontal surface were computed using air temperature, precipitation and potential solar radiation.

4. Results

The stochasticity of the downscaling model allows an infinite number of simulations to be produced—the results presented below were constructed using an ensemble of 20 runs for N_{ds}^c , G_{ds}^c and G_{ds}^f . The other scenarios have only one realisation.

4.1. Evaluation of downscaling model

When calibrated using the S_3^c data set for the WYs 1987–1995 the downscaling model generally explained more than 80% of the variance ($E\%$) in daily temperatures ($TMAX_i$ and $TMIN_i$) and 30–45% in daily wet-day amounts (R_i). As Table 3 indicates the model performance was better during spring (MAM) and autumn (SON) and worse during winter (DJF) and summer (JJA). The relatively low explained variance for the wet-day amounts in all seasons is consistent with previous studies (e.g. Burger, 1996; Wilby et al., 1998a) and underlines

Table 3

Downscaling model calibration “fit” expressed in terms of the percentage of explained variance ($E\%$) in daily R_i , $TMAX_i$ and $TMIN_i$ for the Animas River basin, WYs 1987–1995 (the values for O_i are the percentage of dry and wet days that were correctly assigned by the downscaling model in each season)

Season	Precipitation occurrence (O_i)		Wet-day precipitation amounts (R_i)	Maximum temperature ($TMAX_i$)	Minimum temperature ($TMIN_i$)
	Dry	Wet			
DJF	46	60	45	85	88
MAM	47	67	39	91	92
JJA	54	61	29	79	85
SON	60	59	43	95	96

the difficulty of downscaling local precipitation series from regional (grid-box) scale predictors. At present the unexplained component in daily precipitation amounts must be treated stochastically by the downscaling model.

The $E\%$ statistic is not an appropriate measure of the model fit for precipitation occurrence (O_i) because this predictand is a discrete variable. A more useful metric is the percentage of correct wet and dry day classifications (Wilks, 1995). From Table 3 it is evident that on an average the downscaling model reproduces wet days better than dry days: the success rate for the former was 62% and for the latter 52%. These precipitation statistics were determined from a single realisation of the stochastic downscaling—a severe test of the occurrence model performance. According to the results in Table 3, wet-day occurrence is modelled best in spring and dry-day occurrence in autumn.

Validating the downscaling model using lengthy series of independent data was problematic since the Wolf Creek Pass SNOTEL station has no data prior to 1986. Table 4 compares the downscaling model (N_ds^c) estimates of daily PRCP, TMAX and TMIN

for the WYs 1981–1986 with the observed series S_3^c for the same period. From Table 4 it is evident that the downscaling model produces significantly higher annual precipitation totals, wet-day frequencies and mean wet-day amounts than was recorded at Vallecito Dam and Mineral Creek. Conversely, the downscaling model yielded significantly lower estimates of daily mean TMAX and TMIN, as well as the respective 90th-percentile values than observed. These differences are entirely consistent with what would be expected since the mean elevation of the data shown in Table 4 for S_3^c is 2696 m compared with 2915 m for the N_ds^c .

4.2. Comparison of current rainfall-runoff scenarios

Accordingly, Fig. 5a compares simulated mean daily streamflow by month from the N_ds^c and S_3^c scenarios with observed streamflow values. N_ds^c results are presented as box plots to show the range of simulated mean daily streamflow by month for the ensemble of 20 runs. Fig. 5b shows the percent model error by year for N_ds^c and S_3^c . The modelled hydrographs using the S_3^c consistently show a

Table 4

Comparison of downscaling model output and observed area average PRCP, TMAX and TMIN for the Animas River basin, WYs 1981–1986 (note that the observed statistics do not include data for the SNOTEL site at Wolf Creek Summit which was used in model calibration)

	PRCP (mm)					TMAX (°C)			TMIN (°C)		
	Mean	SD	90%	%Wet	Total	Mean	SD	90%	Mean	SD	90%
Model	5.0	4.6	10.4	62.0	1300	9.3	8.6	20.7	−4.8	7.9	−14.9
Obs	4.6	5.0	10.8	51.4	869	13.9	9.0	26.1	−2.4	8.1	−13.3

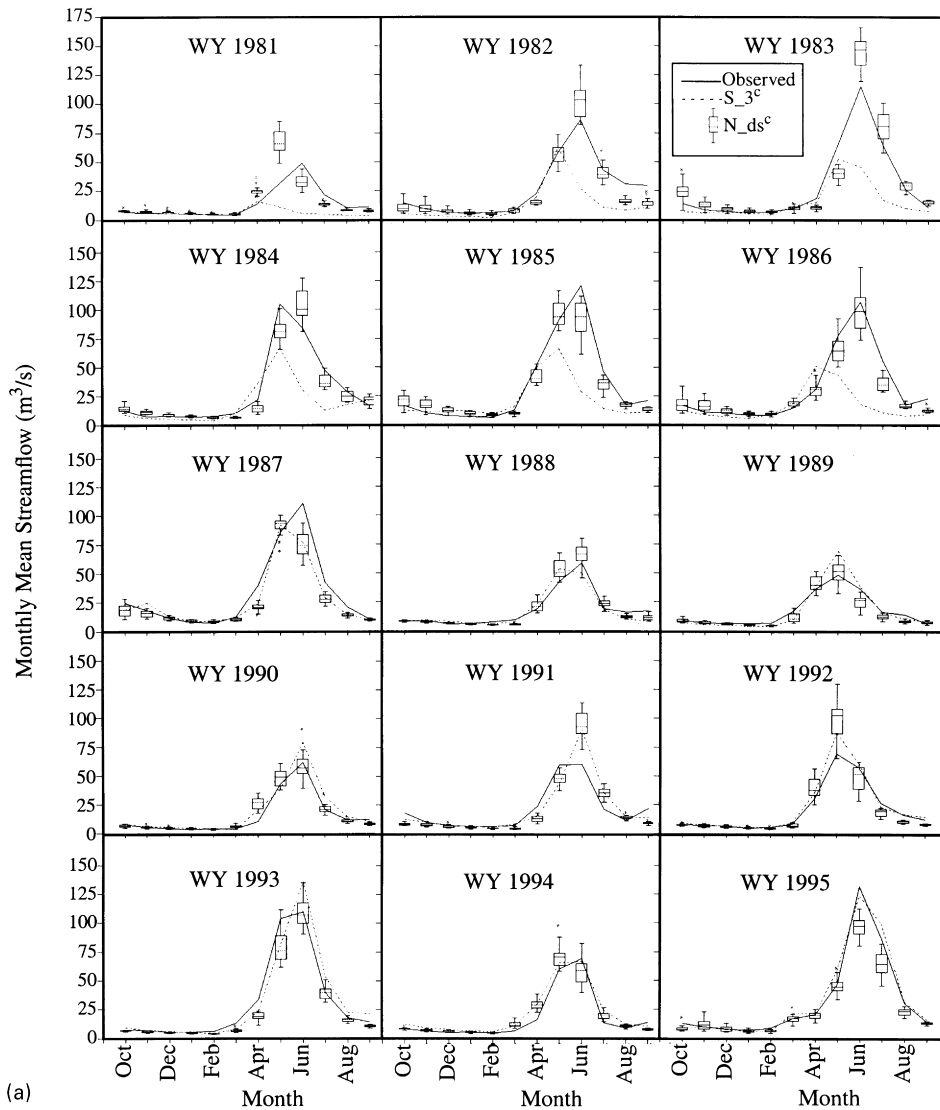


Fig. 5. Comparison of: (a) observed and simulated monthly mean runoff produced by the N_{ds}^c and S₃^c scenarios; (b) percent model error by year for the N_{ds}^c and S₃^c scenarios.

significant underestimation (34–44%) of the gauged runoff for WYs 1981–1986. However, for WYs 1987–1995, errors in S₃^c simulated runoff volumes range from 2 to 12% for all years except 1994 which had an error of 18%.

Fig. 5a and b also provide an independent check of the validity of the downscaling since the statistical models were calibrated using data for WYs 1987–1995. The downscaling model was able to reconstruct

mean daily values for PRCP, TMAX and TMIN prior to 1986 that significantly improved the modelled runoff (relative to the simulation produced by S₃^c) for this period and thereafter the monthly mean of the N_{ds}^c runs is comparable to the results of the S₃^c scenario. These results are taken as a strong endorsement of the downscaling model's capability to reconstruct sub-grid meteorology and hence observed flows (even for an independent period), using just three

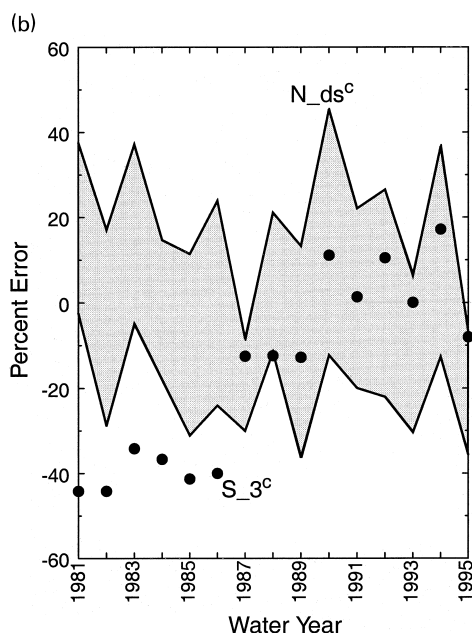


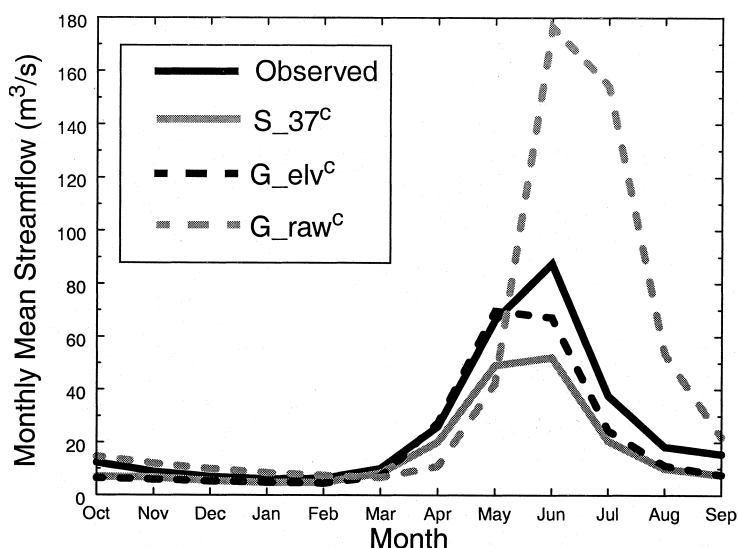
Fig. 5. (continued)

atmospheric predictor variables. They also indicate the potential of the downscaling technique for data reconstruction. Given the problems described for scenario S_{3^c} and the improved performance of the N_{ds^c} scenarios for WYs 1981–1986, the N_{ds^c}

scenarios were assumed to be most representative of the current climatological and hydrological regimes for the purpose of comparisons with all current and future scenarios.

The simulated runoff produced by the two GCM scenarios for the current climate (G_{raw^c} and G_{elv^c}) was evaluated by comparing them with the runoff simulated by S_{37^c} and against observed flows (Fig. 6). As noted previously, scenario S_{37^c} used station data covering the entire San Juan River basin, an area similar in size to a single GCM grid-box. Comparing monthly mean streamflow values in Fig. 6 shows that after redistribution, G_{raw^c} significantly overestimates streamflow from May to September. The G_{elv^c} scenario yielded improved estimates of peak and minimum flows compared with S_{37^c} . The consequences of ‘correcting’ the GCM elevation by month can be seen when comparing the G_{raw^c} and G_{elv^c} scenarios in Fig. 6. The use of higher representative elevations in G_{elv^c} (Fig. 4) results in a drier and warmer regime.

Differences among the current climate scenarios are highlighted by the monthly mean values of PRCP, TMAX and TMIN (Fig. 7a–c, respectively). Each figure shows the range of the monthly mean values for the ensemble runs of N_{ds^c} and G_{ds^c} plus the single realisation of the monthly mean values for G_{elv^c} . Similar trends in PRCP are shown for all

Fig. 6. Simulated monthly mean runoff for observed, S_{37^c} , G_{elv^c} and G_{raw^c} scenarios.

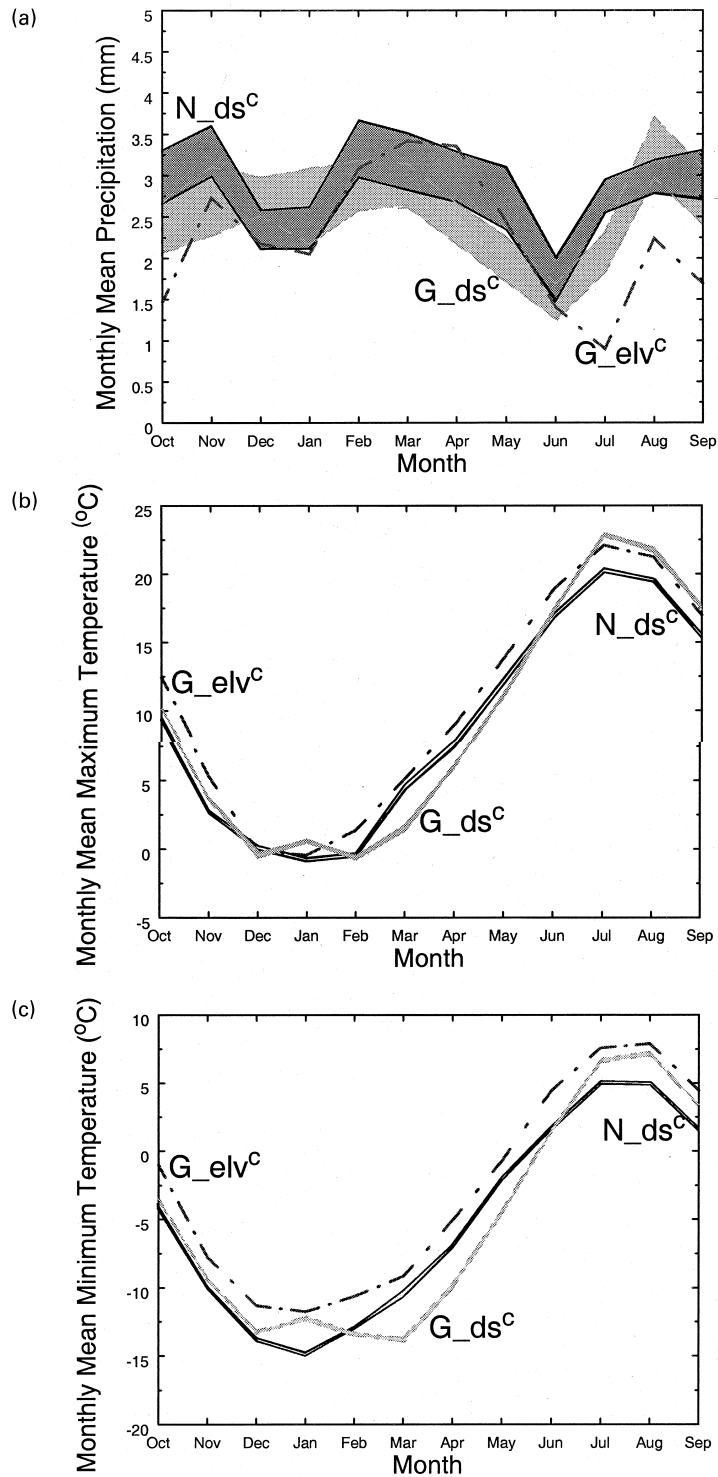


Fig. 7. Monthly mean (a) daily precipitation amounts, (b) maximum and (c) minimum temperatures of the current climate scenarios N_ds^c, S_3^c, G_ds^c and G_elv^c.

Table 5

Changes in daily PRCP, TMAX and TMIN between WYs 1981–1995 and 2081–2095: a comparison of downscaled ($G_{ds}^f - G_{ds}^c$) and HadCM2 ($G_{elv}^f - G_{elv}^c$) scenarios

Season	Change in PRCP (mmd ⁻¹)		Change in TMAX (°C)		Change in TMIN (°C)	
	GCM	Downscaled	GCM	Downscaled	GCM	Downscaled
DJF	+0.73	+1.38	+3.82	+3.35	+5.68	+4.15
MAM	+0.28	+0.02	+0.84	+2.63	+2.33	+2.99
JJA	−0.20	+0.88	+3.16	+3.93	+3.43	+3.94
SON	+0.05	+0.16	+4.87	+3.80	+4.89	+3.43
Annual	+0.21	+0.58	+3.17	+3.43	+4.08	+3.63

scenarios but the magnitudes of the scenarios show large seasonal differences. G_{ds}^c has lower monthly mean PRCP than N_{ds}^c for most of the winter, summer and autumn months while G_{elv}^c has values comparable to N_{ds}^c for the winter to early summer period but less than N_{ds}^c for the remainder of the year. The reasonable agreement between G_{elv}^c and observed streamflow (Fig. 6) reflects the general agreement in winter and spring PRCP between the G_{elv}^c and N_{ds}^c scenarios and the minimal contribution of summer precipitation to streamflow in this region of the United States. Monthly mean TMAX and TMIN for G_{ds}^c and G_{elv}^c are generally warmer than those for N_{ds}^c for all months except those in spring when G_{ds}^c was cooler.

4.3. Comparison of future rainfall-runoff scenarios

Evaluation of the magnitude of the changes between the current and future climates on a seasonal basis after spatial redistribution (Table 5) reveals that the downscaling produced larger increases in summer and winter precipitation than the elevation corrected GCM scenario. Changes in TMAX were greater in the downscaled scenario for spring and the GCM scenario for autumn. Changes in downscaled TMIN were less than those in the GCM for autumn and winter. Such variations, between the downscaled and raw GCM scenarios, are noteworthy given that all scenarios originate from the same GCM.

Fig. 8a and b compare the simulated monthly mean streamflow produced by scenarios N_{ds}^c , G_{ds}^c , G_{ds}^f , G_{elv}^c and G_{elv}^f with observed monthly mean streamflow. For the N_{ds}^c , G_{ds}^c and G_{ds}^f scenarios, the maximum and minimum streamflows arising from the ensemble of 20 runs are given. For

current climate conditions (Fig. 8a), the high level of agreement between N_{ds}^c and observed flows further supports the assumption that scenario N_{ds}^c is representative of the current hydrological regime. G_{ds}^c underestimates observed flows from October to May, overestimates flows in June and provides comparable flows for the remainder of the WY. G_{elv}^c provides reasonable agreement of flows for the period of March–May but underestimates observed flows for all other months. Peak flow occurs in June for N_{ds}^c , G_{ds}^c and observed, while G_{elv}^c peaks one month earlier in May.

For future climate conditions (Fig. 8b), scenarios G_{ds}^f and G_{elv}^f indicate higher winter period flows but lower spring and summer flows than under present climate conditions. No change in peak flow timing from the current climate is indicated for G_{ds}^f but G_{elv}^f peaks one month earlier. Fig. 9 shows the seasonal mean streamflow for all scenarios. There is no significant change in mean annual volume from current to future condition in either G_{ds} or G_{elv} scenarios, but there are differences in the seasonal means.

The reasons for the differences in the monthly flow regimes (Fig. 8a and b) and the similarities in the mean annual volumes (Fig. 9) are to be found in the monthly and seasonal differences in mean PRCP, TMAX and TMIN among the scenarios (Figs. 7a–c and 10a–c). For the future scenarios (G_{ds}^f , G_{elv}^f), warmer autumn and winter TMAX and TMIN values increase the amount of precipitation occurring as rain rather than snow during these months and increase snowmelt rates at lower elevations. These changes result in increased streamflow for these periods. Differences in the magnitudes of the monthly flows reflects the differences in seasonal precipitation type

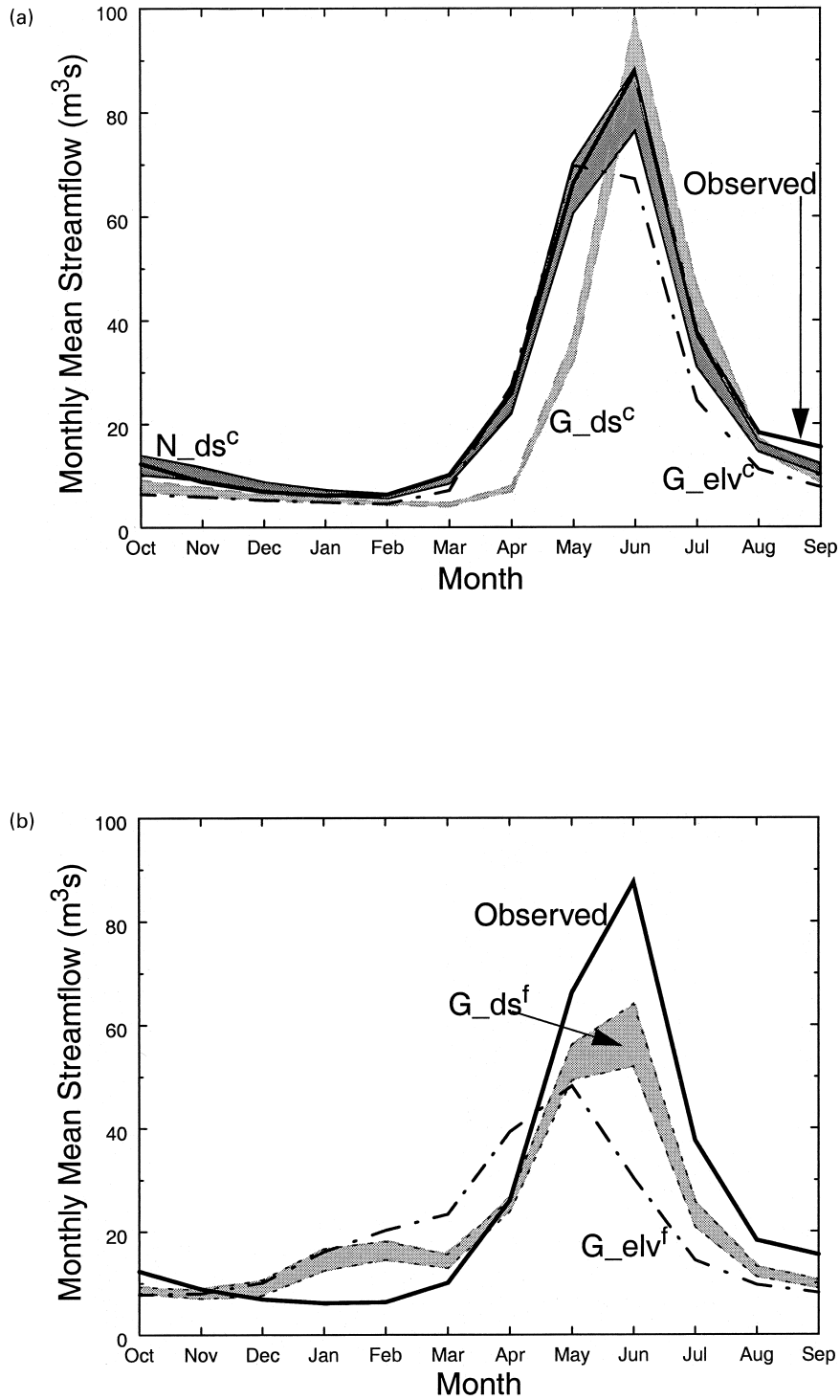


Fig. 8. Monthly mean runoff of the (a) current and (b) future climate scenarios.

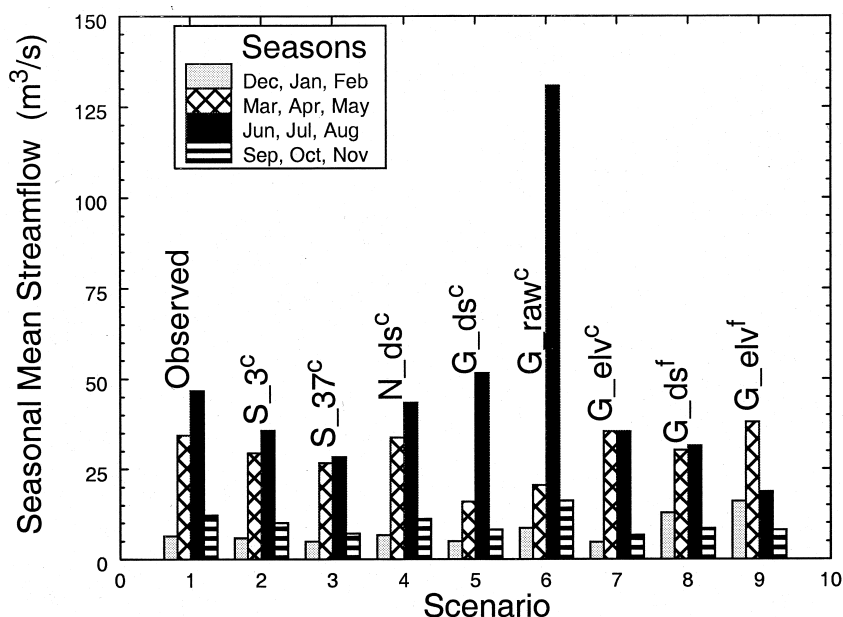


Fig. 9. Mean annual runoff volumes for the current and future climate scenarios.

and amount, as well as the variations in the distribution of TMAX and TMIN.

Changes in the flow regimes (Fig. 8a and b) are also a consequence of associated changes in snowpack behaviour (Figs. 11a and b, and 12a and b). If scenario N_{ds}^c is taken as the datum, it is evident that scenario G_{ds}^c yields snow-covered area durations that are consistent with current conditions (Fig. 11a). In comparison, scenarios G_{ds}^f and G_{elv}^f exhibit significant reductions in the duration of basin snow-covered area (Fig. 11b). All scenarios produce qualitatively similar seasonal regimes of basin snow-covered area: maxima consistently occur in December–February and minima in July–September. Scenarios G_{elv}^c and G_{ds}^f result in maximum snow-covered areas approximately 10% lower than current conditions, compared with as much as 35% lower for scenario G_{elv}^f. (Note that none of the snow-covered area duration curves reach 100% because a different snow area depletion curve was used in PRMS for areas above and below timberline. For areas above timberline, snow covered area is limited to a maximum of 70%. This assumption is made to account for the effects of snow redistribution that normally occurs above timberline).

Comparing snowpack water equivalent relative to the current climate (Fig. 12a) shows comparable values of snowpack water equivalent during the accumulation phase for N_{ds}^c and G_{ds}^c but a delay in melt by 1 month for G_{ds}^c. This delay is expressed in the simulated monthly mean streamflow shown in Fig. 8a. Snowpack water equivalent for scenario G_{elv}^c is less than N_{ds}^c for all months. For future climate conditions (Fig. 12b) scenario G_{elv}^f is by far the most extreme in terms of reduction in snowpack, presumably a consequence of the increases in winter TMAX and TMIN which produce more rain and less snow. Fig. 12b also suggests that there may be subtle changes in the timing of the snowpack accumulation/ablation under future climate conditions. In scenarios N_{ds}^c, G_{ds}^c and G_{elv}^c, peak accumulation occurs in April–May as opposed to March–April in the case of G_{ds}^f and G_{elv}^f. Scenario G_{elv}^f results in significant reductions in total snowpack volume even when compared with the other future climate scenario G_{ds}^f. The later date of peak accumulation and melt for scenario G_{ds}^f, as compared to G_{elv}^f, is consistent with the later streamflow peak shown in Fig. 8b.

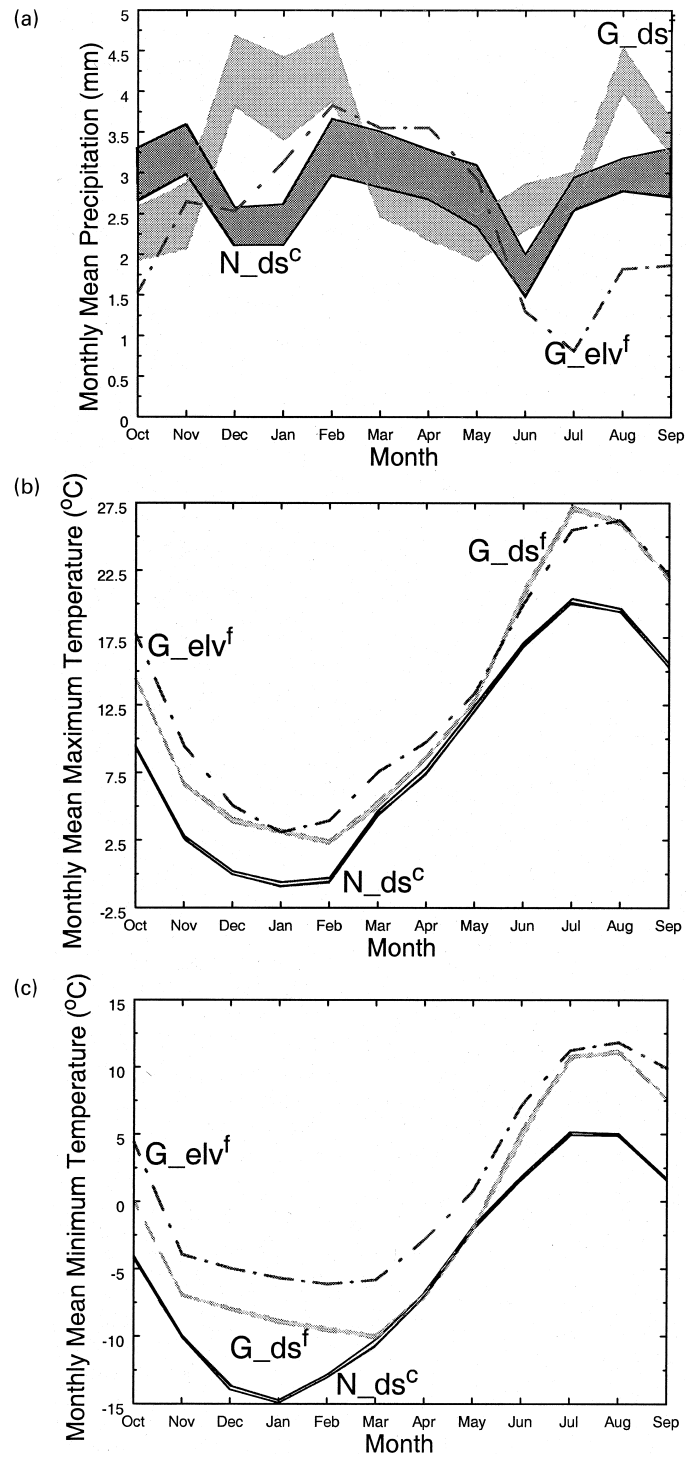


Fig. 10. Monthly mean (a) daily precipitation amounts, (b) maximum and (c) minimum temperatures of N_{ds}^c compared with the future climate scenarios G_{ds}^f and G_{elv}^f .

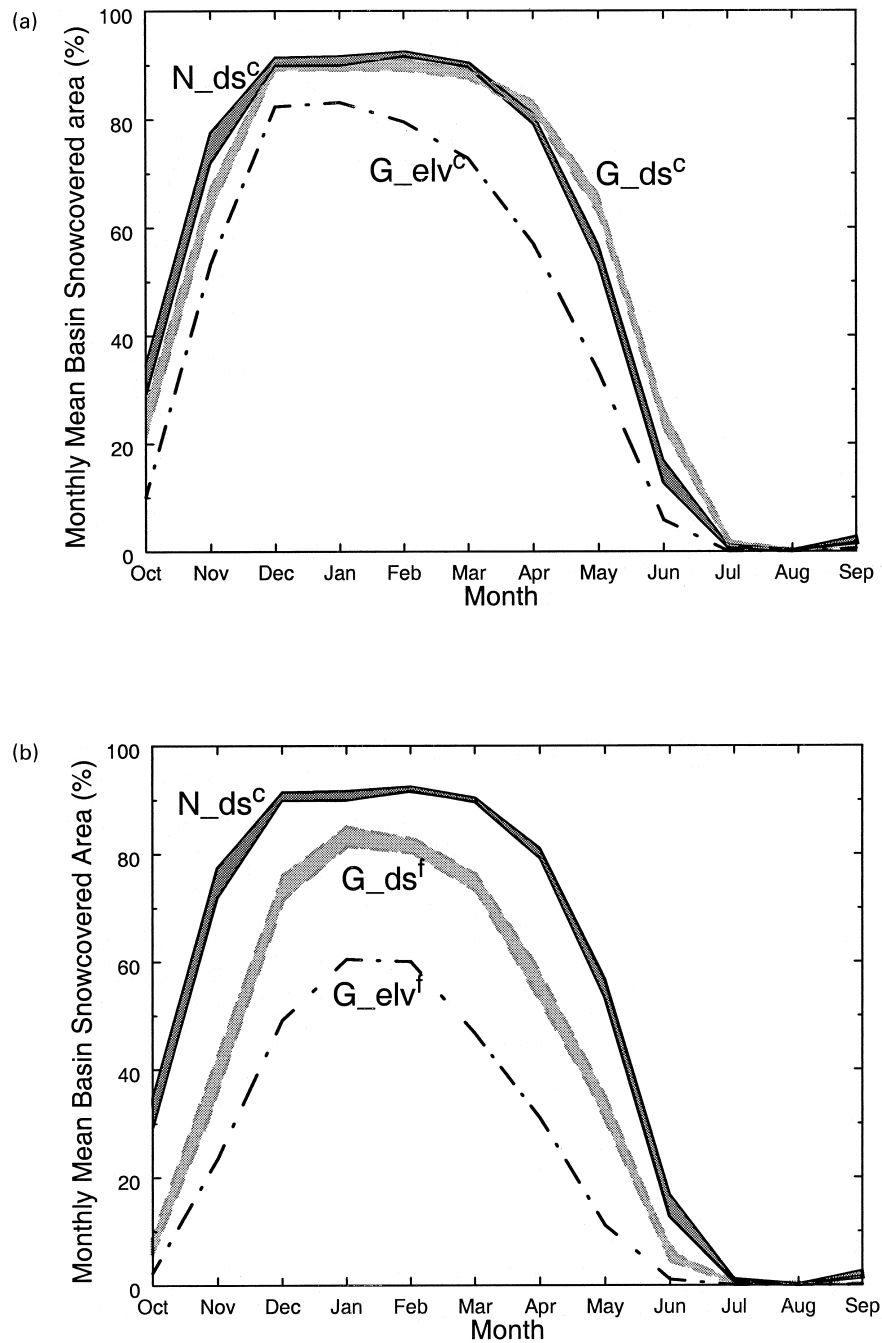


Fig. 11. Monthly mean snow cover areas of the (a) current and (b) future climate scenarios.

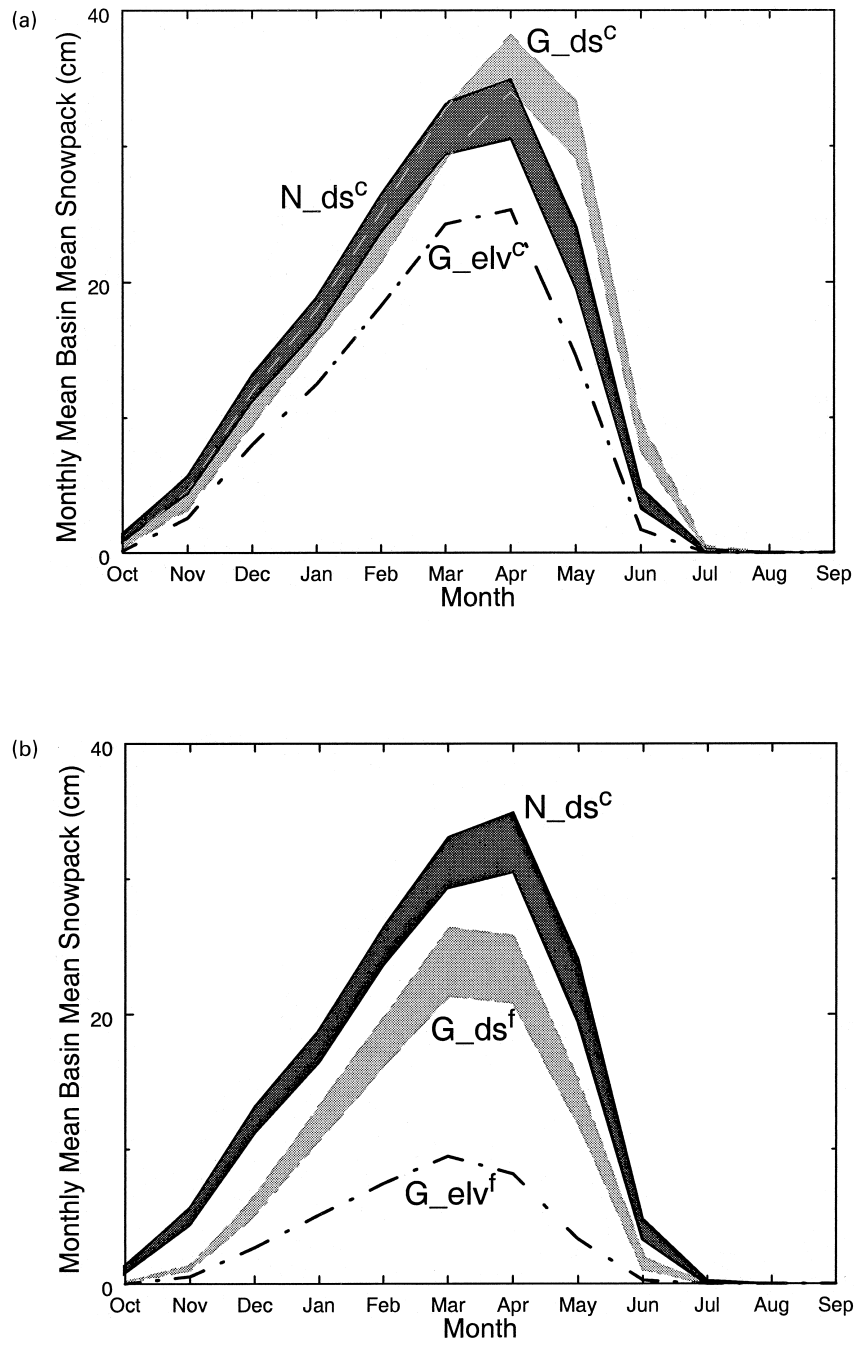


Fig. 12. Monthly mean snowpack water equivalent of the (a) current and (b) future climate scenarios.

5. Discussion

This study compared three methods of generating current and future rainfall-runoff scenarios: (1) statistically downscaled GCM output; (2) raw GCM output; and (3) GCM output corrected for elevational biases. The regression-based statistical downscaling model employed three grid-box predictor variables (daily mean sea level pressure, surface specific humidity and 500 hPa geopotential heights) to simulate sub-grid scale daily precipitation and temperature series. All climate scenarios, whether originating from the downscaling procedure or directly from the GCM output, were spatially distributed across the Animas River basin using lapse rates and topographic information for specified hydrological response units. Finally, a distributed hydrological model (PRMS) was used to simulate daily runoff, snow-covered area and snow-pack under competing GCM-derived and downscaled climate scenarios.

It was demonstrated that downscaled daily precipitation and temperature series for the observed climate (scenario N_{ds}^c) can result in improved simulations of daily runoff for independent periods of record for which the station data are missing or less reliable. The quality of the modelled flows in the Animas River for the period prior to 1986 testifies to the potential of the downscaling as a means of reconstructing local hydrometeorological variables given only grid-box scale atmospheric predictors. In comparison, the three GCM-derived scenarios of the current climate (G_{ds}^c , G_{raw}^c and G_{elv}^c) yielded large seasonal differences in precipitation and generally higher temperatures relative to N_{ds}^c . The correction for elevation biases in the raw GCM output (G_{elv}^c) resulted in significantly improved simulations of the current runoff regime when compared with the uncorrected output (G_{raw}^c).

The distributed hydrological model was next used to compare two future climate scenarios originating from: (1) statistical downscaling (G_{ds}^f); and (2) raw GCM output corrected for elevation biases (G_{elv}^f). Scenario G_{ds}^f yielded modest reductions in summer streamflow and winter snowpack by 2081–2095 relative to current conditions. In comparison G_{elv}^f yielded far greater reductions in modelled flow and snowpack area/water equivalents over the same period. Furthermore, seasonal runoff and snowpack

regimes exhibited marked differences between the two future climate scenarios G_{ds}^f and G_{elv}^f . The timing of the onset of snowpack melt was 1 month earlier for scenario G_{elv}^f than for G_{ds}^f , with concomitant changes in the months of maximum runoff (May and June, respectively).

From these examples it is evident that the choice of technique for scenario generation has major implications for the projected climate change impact (in this case, basin hydrology). So which regional climate change scenario should be used for impact analysis? The statistical downscaling has a number of advantages over the use of raw GCM output. Firstly, the stochasticity of the model facilitates the generation of ensembles of future climate realisations—a pre-requisite to confidence estimation. Secondly, the downscaling model may be tuned to reproduce the unique meteorological characteristics of individual stations—a valuable asset in heterogeneous landscapes or mountainous terrain. Thirdly, such techniques are far less data intensive and computationally demanding than dynamical methods such as nested or regional climate modelling.

The capability of downscaling to reproduce hydrological processes at scales less than a single GCM grid-box is not at issue; what remains uncertain is the extent to which the assumed empirical predictor–predictand relations are valid under future climate conditions (Wilby, 1997). As mentioned previously, downscaled scenarios are sensitive to many factors, including the choice of predictor variables and downscaling domains, season definitions, the chosen mathematical transfer functions and calibration periods (Winkler et al., 1997). For example, Hewitson and Crane (1999) demonstrated that precipitation anomalies over South Africa, downscaled using only atmospheric circulation predictors, were greater and sometimes of opposite sign to those produced via circulation and humidity predictors.

The preceding analyses also indicate that the downscaling model performed better in some seasons than in others. This might be expected a priori given the relatively limited set of predictor variables. For example, the gridded atmospheric predictors were an appropriate means of downscaling sub-grid scale precipitation amounts during winter and spring (when large weather systems dominate the region), but the same predictors were less proficient for summer convective systems (when indices of

atmospheric instability or column integrated moisture content would be more helpful). Unfortunately, the choice of daily predictor variables is ultimately constrained by what has been archived for the GCM experiment under consideration. Until daily moisture variables are routinely archived for multiple levels in the atmosphere it will be necessary to employ other GCM products such as surface humidity, acknowledging that such predictors are often associated with precipitation as both a cause and an effect (Hostetler, personal communication).

There is clearly a considerable scope for further model development and scenario comparison. The statistical downscaling used herein employed proven techniques but it is conceded that different model configurations and sets of predictor variables can yield markedly different regional climate scenarios (see Wilby et al., 1998b). Given the inherent uncertainty associated with, in particular, precipitation downscaling there may be scope for the development of alternative techniques that rely more heavily on high resolution topographic information—data that are currently available in digitised form at higher resolutions than from most remotely sensed or ground-based hydrometeorological observation networks (Beven, 1995). However, even the present downscaling technique has the potential for high resolution real-time flow forecasting in which, for example, the gridded 0–14 day 850 hPa temperature and 0–14 day accumulated precipitation ensemble forecasts of the Environmental Modeling Center's (NCEP) Medium Range Forecast Model might be used to downscale station-scale PRCP, TMAX and TMIN as input to PRMS.

In the meantime, for the Animas River basin, there is at least a qualitative consensus amongst the models that the future magnitude of low and intermediate flows will increase by 2081–2095, whereas peak flows, basin snow-covered areas and snowpack water equivalents will all decline relative to current conditions. Research is ongoing to determine the generality of these water resource impacts for other regions, GCM outputs and downscaled scenarios (e.g. Hay et al., 1999).

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